

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 13A.4

FACTORS AFFECTING EQUILIBRIUM CONSTANTS 2

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A.4: Factors Affecting Equilibrium Constants 2
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	7 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 13A.4

Q1. 1. Explain, in terms of K_p , why an increase in total pressure increases the yield of ammonia in the reaction:
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

[4 Marks]
WCH14/01/Jun23/Q5

Q2. 2. What is the effect of adding a catalyst on the value of K_c ?

[1 Marks]
WCH14/01/Jan22/Q5

- A. K_c increases
- B. K_c decreases
- C. K_c remains unchanged
- D. K_c is halved

Q3. 3. A system at equilibrium is disturbed by increasing the concentration of a reactant. State and explain the effect on the value of K_c .

[2 Marks]
WCH14/01/Oct21/Q4

SECTION B: WORKED MARK SCHEME — 13A.4

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Explain, in terms of K_p , why an increase in total pressure increases the yield of ammonia in the reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ 1. K_p is unchanged by pressure. 2. When pressure increases, the ratio $p(NH_3)^2 / (p(N_2) * p(H_2)^3)$ decreases because the denominator has more moles of gas (4 vs 2). 3. To restore K_p , $p(NH_3)$ must increase and $p(N_2)$, $p(H_2)$ must decrease. 4. Hence equilibrium shifts to the right, increasing the yield.	[4]
Q2	Q2: 2. What is the effect of adding a catalyst on the value of K_c ? C	[1]
Q3	Q3: 3. A system at equilibrium is disturbed by increasing the concentration of a reactant. State and explain the effect on the value of K_c . 1. K_c remains unchanged. 2. K_c is only affected by temperature, not by changes in concentration.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 13A.4

FAQ #1: How does pressure affect the yield if K_p is constant?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How does pressure affect the yield if K_p is constant?

MODEL ANSWER / EXAMINER EXPECTATION:

When pressure changes, the partial pressures change. This alters the quotient (the ratio of products to reactants). The system then shifts to bring the quotient back to the constant value of K_p .

FAQ #2: Why does a catalyst not affect the equilibrium constant?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why does a catalyst not affect the equilibrium constant?

MODEL ANSWER / EXAMINER EXPECTATION:

A catalyst speeds up both the forward and reverse reactions equally. It helps the system reach equilibrium faster but does not change the position of equilibrium or the ratio of products to reactants at equilibrium.